III. MEASUREMENTS

A. Equipment

Several sound-measuring instruments are available to CSHOs. These include sound level meters, noise dosimeters, and octave band analyzers. This section describes general equipment care, followed by the uses and limitations of each kind of instrument.

1. Noise Evaluation Instrument Care and Calibration

Instruments that measure noise contain delicate electronics and require practical care. Store and transport the equipment in its custom case. Be aware of the instrument manufacturer's recommendations for proper storage (for example, some manufacturers recommend removing all batteries from stored equipment, while others require a primary battery to remain in the instrument). Make sure batteries will last the anticipated sampling period. A battery tester can be useful. CSHOs may need to install fresh batteries or recharge reusable batteries with a battery charger.

> *All instruments must be calibrated (according to the manufacturer's instructions) to ensure measurement accuracy. [29 CFR 1910.95(d)(2)(ii)]*

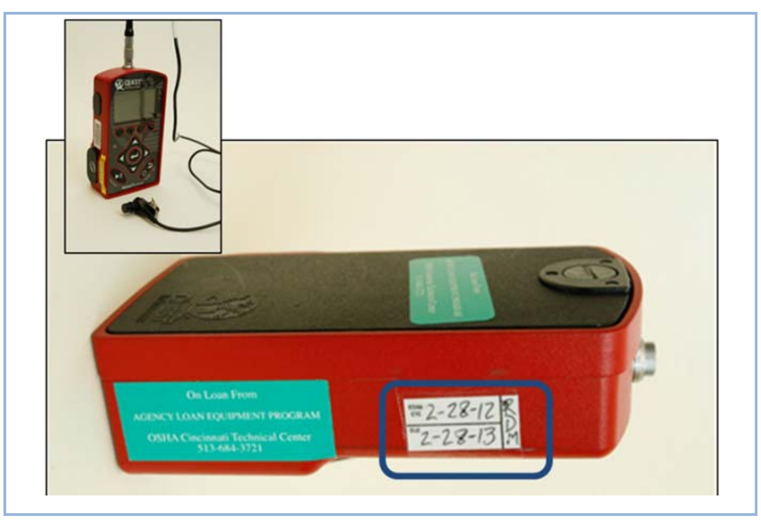
All noise-measuring instruments used by CSHOs require two types of calibration:

- Periodic factory-level calibration (e.g., annual)
- Pre- and post-use calibration

Both pre- and post-inspection calibrations are required for any noise instruments used by CSHOs. It is important to understand the difference between these two types of calibrations. Calibrators must also be calibrated on an annual basis.

Equipment manufacturers typically recommend ***periodic calibration*** on an annual basis. These rigorous testing protocols ensure that the electronic components are in good working order and detect shifts in performance that indicate gradual deterioration. Periodic calibration results in a calibration certificate documenting the standard of performance. Typically, the instrument will also receive a sticker indicating its last calibration date and when the next periodic calibration is due (Figure 12). An instrument owned by OSHA that is past its calibration due date must be returned to OSHA's Cincinnati Technical Center (CTC) to have its calibration renewed. Do not continue to use it past the calibration date.

Figure 12. Noise Dosimeter Calibration Sticker



During periodic calibration, the CTC also performs preventive maintenance to ensure that the equipment remains fully functional over its life expectancy. If the calibration team detects a problem, it services the instrument as necessary. When returning equipment to CTC for periodic calibration, be sure to include a note about any problems or concerns with equipment function so they can be evaluated as part of the maintenance process. If equipment is not functioning well, CTC requests that the instrument be returned for inspection, even if it is not yet due for calibration.

OSHA's CTC is qualified to perform periodic (annual) calibration for the noise-monitoring instruments commonly issued to CSHOs. CTC also coordinates periodic factory calibration of any OSHA-owned noise-monitoring instruments that it does not service directly.

Employers that lease or own Type I or Type II noise-measuring instruments can arrange annual calibration of the equipment through the equipment supplier or manufacturer.

Octave band analyzers that are integrated into a sound level meter will be calibrated as part of the sound level meter. However, detachable octave band analyzers must be returned to CTC for periodic calibration with the meter with which they are intended to be used

Pre- and post-calibration procedures confirm that the instrument is functioning properly on the day that it is used and prove that it is still registering sound levels correctly at the end of the day. Pre- and post-calibrations also confirm that changes in temperature or humidity have not affected the instrument's accuracy. If practical, spot check the instrument with a calibrator after the stabilization period.

When unpacking a cold instrument in a warm environment, or moving from one temperature zone to another, allow the instrument at least 5 minutes to stabilize for each 18˚F (10˚C) of change.

Each instrument model is calibrated in a slightly different manner, but the general process follows basic standard steps. Typical daily ***pre-use calibration*** involves (1) setting up the instrument for use, (2) turning on both the electronic "calibrator" and the noise-measuring instruments to allow them to "warm up," (3) checking the calibrator and instrument battery charge, (4) testing the instruments with a standard tone of known pitch and intensity produced by the calibrator (e.g., 114 dB at 1,000 Hz), (5) checking the instrument reading during the test and making minor adjustments to the instrument if necessary, and (6) documenting the calibration results. For the ***post-use calibration check,*** the process is repeated, without step 5, after the instrument has been used. Both the pre- and post-use calibration must be documented (If it isn't properly documented, it didn't happen). See Figures 13 and 14 for illustrations of this process for dosimeters and sound level meters, respectively.

Figure 13. Noise Dosimeter Calibration

Figure 14. Sound Level Meter Calibration

Confirm that you understand the procedures for calibrating each of the instruments you use. If in doubt, review instructions in each instrument's user's manual and consult CTC if questions arise. In general, as long as the sound level readout is within 0.2 dB of the known source (the calibrator output), it is suggested that no calibration adjustments be made. If large fluctuations (greater than 1 dB) in the level occur, then either the calibrator or the instrument may have a problem.

Review your noise instrument calibration procedure and check whether your process:

1. *Confirms that both the calibrator and the instrument have not exceeded the periodic calibration due date.*
2. *Uses the correct calibrator for the instrument.*
3. *Uses the correct adaptor between the calibrator and the instrument microphone.*
4. *Confirms the battery charge.*
5. *Adjusts the instrument calibration when the tolerance is within the manufacturer's published limits (e.g., ±0.2 to 1 dB) but rejects the equipment if the calibration reading is outside the limits (e.g., ±1 dB or more).*
6. *Prevents use of equipment that is outside its periodic calibration due date or fails pre-use calibration.*
7. *Creates a record of pre-use calibration.*

Additionally, confirm that you know how to change the battery in both the calibrator and the instruments. If in doubt, review instructions in each instrument's user's manual. A low battery is the number-one cause of equipment failing pre- and post-use calibration. Changing the battery will often bring the equipment back into an acceptable calibration range immediately, but a little practice is needed to change the battery quickly on some equipment. Be prepared, so that a low battery doesn't slow you down during an early morning calibration session (Figure 15).

Figure 15. Changing Equipment Batteries

Noise measurements collected by CSHOs cannot be used as a basis for citations unless they are obtained using equipment that has a current (within the past 12 months) periodic calibration certificate on file and that has received ***documented*** calibration before and after the measurements were made using accepted practices for documentation, as outlined in the OSHA *Field Operations Manual*.

2. *<u>Sound Level Meters</u>*

Sound level meters provide instantaneous noise measurements for screening purposes (Figure 16). During an initial walkaround, a sound level meter helps identify areas with elevated noise levels where full-shift noise dosimetry should be performed. Sound level meters are useful for:

- Spot-checking noise dosimeter performance.
- Determining a worker's noise dose whenever a noise dosimeter is unavailable or inappropriate.
- Identifying and evaluating individual noise sources for abatement purposes.
- Aiding in engineering control feasibility analysis for individual noise sources being considered for abatement.
- Evaluating the suitability of HPDs for the actual noise level in an area.

Figure 16. Sound Level Meter

i) Sound Level Meter Types and Performance

Sound level meters used by OSHA meet American National Standards Institute (ANSI) Standard S1.4-1971 (R1976) or S1.4-1983, "Specifications for Sound Level Meters." These ANSI standards set performance and accuracy tolerances according to three levels of precision: Types 0, 1, and 2.

- Type 0 is used in laboratories.
- Type 1 is used for precision measurements in the field.
- Type 2 is used for general purpose measurements.

The most widely used sound level meter for workplace evaluations, the ***Type 2 meter,*** performs with the minimum level of precision required by OSHA for noise measurements. These meters are usually sufficient for general purpose noise surveys. For compliance purposes, readings obtained with a Type 2 sound level meter are considered to have an accuracy of ±2 dBA.

In contrast, a ***Type 1 meter*** has an accuracy of ±1 dBA. The Type 1 meter accuracy, precision, and additional features make it the preferred model for obtaining readings that will be used to help design cost-effective noise controls.

For unusual measurement situations, refer to the manufacturer's instructions and appropriate ANSI standards for guidance in interpreting instrument accuracy.

Other types of sound level meters also exist but do not meet ANSI requirements for the Type 2 or Type 1 designation. These meters, which are often modestly priced, can be useful pre-screening tools for employers seeking to identify noisy locations and track improvements during noise reduction efforts. They cannot, however, be used to document compliance with OSHA standards; only properly calibrated Type 2 or Type 1 meters can

One model of sound level meter typically used by CSHOs, the Quest SoundPro, is designed to operate in temperatures of 14˚ to 122˚F (-10˚C to 50˚C).

Over this range, temperature has a modest effect on the accuracy of measurements (less than ±0.5 dB). Likewise, the sound level meter can be expected to operate effectively between 10% and 90% relative humidity.

serve that purpose. For example, sound level meter applications are available for some smartphones. Such an application can give a rough estimate of the noise level in a particular location but may not be used to document compliance with OSHA standards.

All sound level meters are affected by temperature and humidity; however, these instruments are intended to provide reliable readings within the normal range of workplace temperatures. During extreme weather, temperatures might be considerably outside that range in untempered storage (e.g., the trunk of a parked car). Avoid storing noise measurement equipment where the temperature could be lower than -13˚F (-25˚C) or higher than 158˚F (70˚C). Avoid carrying cold equipment into a very humid environment, which could permit moisture to condense on the instrument. To prevent this situation, do not keep noise equipment in the trunk of a cold car; instead, carry it in the passenger compartment and store it indoors at the destination. For equipment that must be carried for a brief time into a very cold area to collect a measurement, one strategy is to keep the equipment under a coat (or otherwise wrapped/insulated), if possible, to keep it from getting cold.

Sound level meters should be calibrated using the steps outlined in Section 1, above, and according to the manufacturer's instructions.

ii) <u>Using a Sound Level Meter</u>

Different work environments and different sound level meter microphones might require variations in measurement procedures. For practical purposes, however, certain basic steps apply in most circumstances.

Confirm that the sound level meter is properly calibrated and temperature-stabilized. Then, position the microphone in the monitored worker's hearing zone. OSHA defines the hearing zone as a 2-foot-wide sphere surrounding the head. Considerations of practicality and safety will dictate the actual microphone placement at each survey location. Note that when noise levels at a worker's two ears are different, the higher level must be sampled for compliance determinations.

Figure 17. Sound Level Meter Positioning

Keep in mind that your body or surrounding equipment can influence the noise level, acting as a barrier between the noise source and the microphone. Hold the sound level meter away from your body to minimize this effect (Figure 17).

Consult the manufacturer for any specific instructions for positioning the model of sound level meter you plan to use. This may be particularly important when measuring in unusual settings. For example, the manufacturer may have specific instructions for sound level readings in a non-reverberant environment.

Use a wind screen to reduce measurement errors caused by wind turbulence over the microphone. Typical wind screens are made of soft foam rubber and are designed to fit over the microphone (Figure 18). Although not necessarily needed indoors if air movement is minimal, a wind screen can be left in place for all measurements. Collected measurements can be affected by anything that comes across the face of the sound level meter microphone, such as hair, shirt collars, scarves, or other objects. The use of a wind screen reduces the effects of this incidental contact. Wind screens have the added advantage of protecting the microphone, at least somewhat, from damage resulting from impact, dust, paint overspray, and moisture.

Figure 18. Wind Screen

Most Type 1 and Type 2 sound level meters can be set to respond with either a “slow response” or a “fast response”. The meter dynamics are such that the meter will reach 63% of the final steady-state reading within one time constant:

- ***Fast response*** corresponds to a 125-millisecond (ms) time constant.
- ***Slow response*** corresponds to a 1-second time constant.

The meter screen shows the average sound pressure level measured by the meter during the period selected. In most industrial settings, the meter fluctuates less (and therefore is easier to read) when measurements are made with the slow response rather than the fast response. A rapidly fluctuating sound generally yields higher maximum sound pressure levels when measured with a fast response. The choice of meter response depends on the type of noise being measured, the intended use of the measurements, and the specifications of any applicable standards. For typical occupational noise measurements, including extremely elevated short-term noise (e.g., noise that will be compared to the 115 dBA maximum for a 15-minute period), the meter response on a sound level meter should be set at slow. For more information on OSHA’s standard for extremely elevated short-term noise exposures see Section II.I.2— OSHA Noise Standards.

Many sound level meters also have “peak” and “impulse” response settings for measuring transient sounds (sounds that decay or pass with time). These settings are not interchangeable; the true peak value is the maximum value of the noise waveform, while the impulse measurement is an integrated measurement. It is appropriate to use the true peak reading only when determining compliance with OSHA's 140-dB peak (instantaneous) sound pressure level [29 CFR 1910.95(b)(1) or 29 CFR 1926.52(e)]. Avoid using the impulse response setting when measuring true peak sound pressure levels.

Note that noise dosimeters and sound level meters that are set to integrate or average sound over a period of time do not use either the fast or slow time constant; they will sample many times per second.

3. <u>*Octave Band Analyzer*</u>

Most sounds are not a pure tone but rather a mix of several frequencies. The frequency of a sound influences the extent to which different materials attenuate that sound. Knowing the component frequencies of the sound can help determine the materials and designs that will provide the greatest noise reduction. Therefore, octave band analyzers can be used to help determine the feasibility of controls for individual noise sources for abatement purposes and to evaluate whether hearing protectors provide adequate protection.

i) <u>Octave Band Analyzer Types and Performance</u>

Octave band analyzers segment noise into its component parts. The standard octave band filter set provides filters with the following center frequencies: 16; 31.5; 63; 125; 250; 500; 1,000; 2,000; 4,000; 8,000; and 16,000 Hz. The special signature of a given noise can be obtained by taking sound level meter readings at each of these settings (assuming that the noise is fairly constant over time). The results may identify the octave bands that contain the majority of the total radiated sound power (Figure 19).

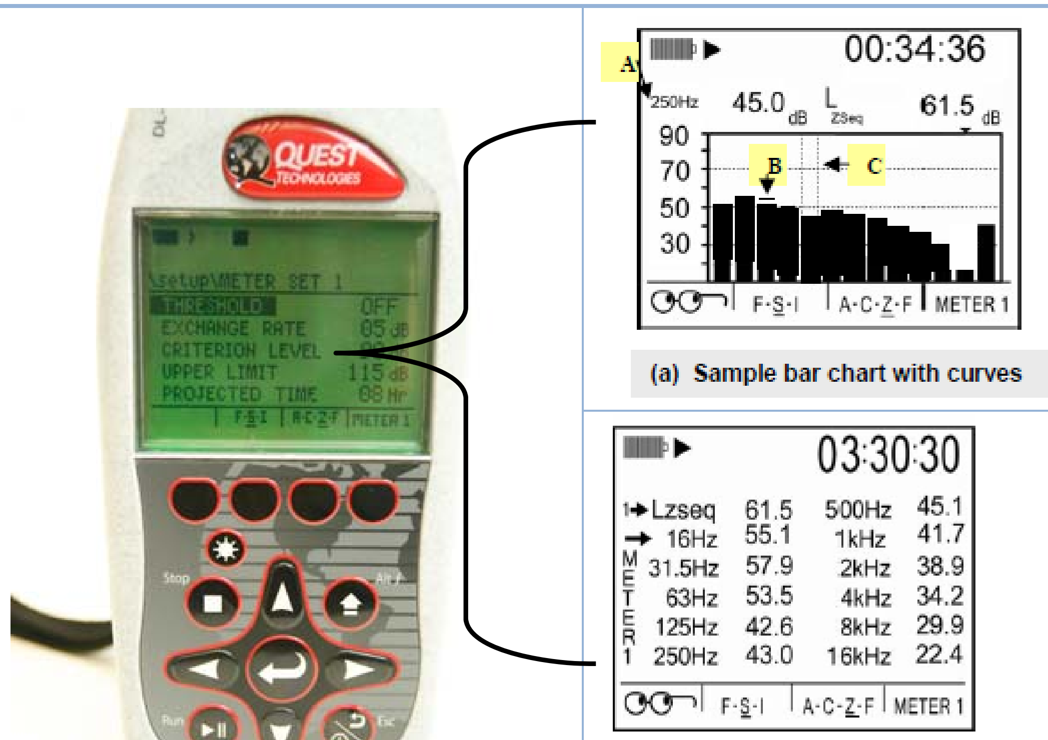
Question: *I've heard that some sound level meters should be pointed at the noise source, while others should be held at an angle (e.g., 70 degrees, 90 degrees).*

Answer: *In many cases, orientation makes no significant difference, but it is always best to follow any recommendation from the manufacturer. Such a recommendation would be based on microphone type. Typical recommendations include:*

- *Free-field microphones—point directly toward the noise source (a 0-degree angle).*
- *Random incidence microphones—hold at a 70-degree angle to the source.*
- *Pressure microphones—hold at a 90-degree angle to the source.*

CSHOs should consult with CTC regarding the microphone models provided with their sound level meters.

Figure 19. Octave Band Analyzer
Settings and Center Frequencies



(a) Sample bar chart with curves

(b) Tabulation screen

Press "Enter" (center arrow) key to switch screens.

Sample bar chart screen: (A) selected frequency band (250 Hz in example), (B) selected frequency in curve, (C) amplitude (dB) in band.
Tabulation screen: lists amplitude in dB for each frequency band.

For octave band analysis, the ideal sound level meter network (weighting) scale setting is one that provides no weighting at all, such as the Z-weighted scale, which has an unweighted flat response across the entire frequency spectrum from 10 Hz to 20,000 Hz. The C-weighted scale is also an acceptable option for octave band analysis because, in the range of most workplace noise level measurements, unweighted sound level measurements are less than 1 dB higher than the corresponding C-scale measurements. The A-weighted scale, however, is not an appropriate setting for octave band analysis because, by definition, it influences the meter response differently at various frequencies in the range of normal human hearing.

For a more detailed analysis, the spectrum is sometimes measured in one-third octave bands. Although one-third octave bands can be useful for noise engineers concerned with precise frequency measurements, the standard single octave bands are sufficient for most evaluations performed by OSHA.

Whether detachable or integrated into a sound level meter, an octave band analyzer receives its daily calibration in conjunction with the sound level meter with which it will be used. This might involve activating an additional setting during the daily meter calibration. Consult the user's manual for the equipment you will be using.

ii) <u>Using the Octave Band Analyzer</u>

The Type 1 sound level meters used by OSHA (such as the Quest SoundPro) have built-in octave band analysis capability. Some other models of sound level meter are designed to work with a separate octave band analyzer that is physically attached to the meter (Figure 20). In either case, the sound level meter microphone operates normally, but the noise signal detected by the microphone is separated into its component frequencies. When the octave band analyzer is activated and a particular frequency band selected, the meter readout provides the decibel level associated with that frequency. By sequentially switching the meter to each frequency band and taking a reading, the CSHO can determine which octave bands are most represented in the noise.

For example, an octave band analysis providing the following results indicated that the frequencies around 500 Hz and 1,000 Hz were most prominent (Table III–1):

Table III–1. Octave Band Analysis (Noise A)

Hz	31.5	63	125	250	**500**	**1,000**	2,000	4,000	8,000	16,000
dB	68	69	72	76	**89**	**92**	74	77	71	71

In contrast, the following octave band analysis (Table III–2) obtained during concrete demolition (multiple noise sources) indicated that nearly all frequencies contributed to the noise level at that position—a distance of 60 feet from the demolition point. At that point, the overall sound level was 91 dB, demonstrating a standard principle of sound: the sum of all octave bands is greater than any single octave band reading, but the logarithmic values cannot be summed by simple arithmetic addition. See Appendix B for more information on determining the sum of two or more sound levels.

Table III–2. Octave Band Analysis (Noise B)

Hz	31.5	63	125	250	500	1,000	2,000	4,000	8,000	16,000
dB	81	87	83	83	83	86	86	87	82	68

Figure 20. Octave Band Analyzer Graph

Some octave band analyzers can be set to automatic function (i.e., the instrument automatically checks the sound level of each frequency band and stores the results). Other instruments require the user to manually switch between the different frequency bands, recording each reading in sequence.

Variable frequency sounds and sounds that constantly vary in intensity present a challenge to frequency analysis. Unless the sound is relatively constant throughout the process of evaluating all frequency bands, it might not be possible to obtain an accurate reading. The CSHO should attempt to determine whether cyclic sounds have a stable period during which readings would be more accurate.

4. <u>Noise Dosimeter</u>

Like a sound level meter, a noise dosimeter can measure sound levels. However, the dosimeter is actually worn by the worker to determine the personal noise dose during the workshift or sampling period (Figure 21). Noise dosimetry is a form of personal sampling, averaging noise exposure over time and reporting results such as a TWA exposure or a percentage of the PEL.

Dosimeters can be used to:

- Make compliance measurements according to OSHA's Noise standard.
- Measure the worker's exposure to noise over a period of time (e.g., a task or an entire workshift) and automatically compute the necessary noise dose calculations.

Increasingly, some sound level meters can function as noise dosimeters (although they are larger than typical dosimeters), while many noise dosimeters provide instantaneous sound level readings in decibels and therefore can be used as Type 2 sound level meters.

Figure 21. Noise Dosimeter

i) Noise Dosimeter Types and Performance

Most noise dosimeters operate with the precision and accuracy of a Type 2 sound level meter. Therefore, the variations in dosimeter types are primarily a function of either the physical form or the analytical features of each model. Historically, the typical noise dosimeter has included a small positionable microphone connected to the dosimeter by a thin cable. The microphone sits in the worker's hearing zone (e.g., shoulder or lapel near the ear), while the dosimeter clips to the worker's belt. Advances in miniature electronics and wireless technology, however, have permitted manufacturers to offer similar capabilities in a wider range of physical forms (e.g., wireless microphones that clip to the worker's shoulder and transmit information back to a base station, miniature microphones that measure sound levels in the worker's ear).

Function also varies. Simple dosimeters record a single channel and report basic dosimetry results. More complex models can record as if they were three or four separate dosimeters, each integrating the sound level over time using different criteria (e.g., 3 dB and 5 dB exchange rates, different threshold settings).

Noise dosimeters are subject to the same sensitivity to temperature and humidity as sound level meters. Although some have water-resistant housings, they should still be treated as sensitive electronic instruments and be protected from moisture and physical impact. The dosimeter calibration process is nearly identical to that for sound level meters. Frequently, for a given brand of instruments, the same calibrator can be used for a manufacturer's sound level meters and noise dosimeters (Figure 22).

Figure 22. Calibrator Adapter

Always consider the accuracy of noise-measuring equipment when using readings for compliance purposes.

Like Type 2 sound level meters, Type 2 noise dosimeters have an implied accuracy of ±2 dBA. To prove an overexposure, the 8-hour TWA sound level (L-TWA), must be 2 dBA over the PEL.

In practice, the workers are overexposed to noise with an 8-hour TWA of 92 dBA (a dose of 132% as measured at the 90-dBA threshold setting of the dosimeter) and an average sound level of 92 dBA.

Workers must be included in a hearing conservation program when measured noise levels are 87 dBA as an 8-hour TWA (a dose of 66% of the PEL as measured at the 80-dBA threshold setting).

Noise dosimeters routinely must run for 8 to 10 hours per day. This means battery function is particularly important. Some models might require new batteries each day of use. Just as for sound level meters, each dosimeter must receive periodic calibration every 12 months and a daily calibration and battery check before each use. They also require a post-use calibration check. The documentation procedures are the same as those for sound level meters.

ii) Using Noise Dosimeters

According to OSHA's Noise standard (29 CFR 1910.95), the noise dosimeter is the primary instrument for making compliance measurements. Before use, the dosimeter must be set up to record noise exposure using the following criteria:

- Exchange rate: 5 dB
- Frequency weighting: A
- Response: slow
- Criterion level: 85 dBA (Hearing Conservation) or 90 dBA (Administrative and Engineering Controls).
- Threshold: 80 dBA (Hearing Conservation) or 90 dBA (Administrative and Engineering Controls).

As noted above, some dosimeters can simultaneously record exposure using two sets of criteria. With these instruments, the CSHO can obtain separate noise exposure levels based on both the 80 dBA and the 90 dBA threshold. Other noise dosimeters that lack this feature must be set to record using one of these thresholds or the other.

In addition to the 8-hour TWAs, OSHA's noise standards list a short-term level of 115 dBA for a 15 minute period, which is not to be exceeded; this is for steady state sounds measured on the slow response setting. Although sound this loud is unusual, some dosimeter models indicate when the maximum allowable sound level of 115 dBA has been exceeded. This signal should not be used for compliance determination, however, because it might not take the duration of the exposure to this noise level into consideration. But noise that exceeds 115 dBA should be incorporated into the overall TWA noise exposure determination (see Section II.I.2—OSHA Noise Standards for more information). The standard for short-term noise levels is distinct from OSHA's instantaneous ceiling limit of 140 dBA for impact noises (occurring less frequently than one per second and typically measured using a sound level meter set to the fast response setting.

You will need to make other decisions regarding dosimeter setup. For example, the typical noise dosimeter offers several options for the frequency with which noise is sampled and data logged. The more frequently the data are logged, the more data points are stored (and the larger the file eventually will be).

The calibrated noise dosimeter fastens to the worker's belt, while the microphone clips to the shoulder or lapel. Orient the microphone so it points straight up—you might need to adjust the clip to find a functional position. Avoid positioning the microphone where it could become enfolded in clothing or rub against cloth or other materials, both of which could influence the results.

If appropriate, run the microphone cable under the worker's outer layer of clothing to keep it out of the way and prevent it from snagging on objects in the work area. The dosimeter can hang inside the outer layer of clothes as well (an advantage in wet weather), but the microphone must remain in the open air without contacting other surfaces (except the base on which it clips).

Some dosimeter models are capable of taking separate measurements (studies) for different job tasks or processes within the same workshift. The dosimeter can isolate the loudest job task the worker performs. This data can be reviewed later by the CSHO to determine which job tasks contributed most to a worker's overall 8-hour TWA. This feature is useful for assessing engineering controls.

The dosimeter microphone must be protected from wind and harsh materials. Wind screens are optional indoors if air currents are minimal. Always use a windscreen in areas with air motion, outdoors, and in dusty locations or during jobs when the microphone might get dirty (Figure 23). The foam rubber wind screen will help protect the microphone. Additional precautions are required to protect the microphone under the particularly harsh conditions that occur during abrasive blasting, when the microphone should be clipped inside the abrasive blasting helmet.

Workers are understandably curious about the noise dosimeter, and particularly the microphone. Take time to explain that it only collects information on how loud the sounds are—it does not record speech. Activate the dosimeter and replace its screen cover, or lock out the controls before the worker begins working. As a good practice, take sound level measurements frequently during the course of the noise dosimetry. The sound level measurements document the noise in the area at specific points in time and from specific sources. These values both validate the dosimeter reading and provide insight into how and when exposure is occurring. Some noise dosimeters log data that can be downloaded to a computer and later graphed against time to show how the worker's noise exposure varies over the course of a shift. This is a useful feature, but is not a substitute for good notes on the workplace and the sources of noise in specific times and places.

Figure 23. Microphone Positioning and Wind Screen Use

OSHA's Health Response Team (HRT) maintains the following specialized noise analysis equipment, which can be used for noise exposure and engineering control evaluations:

Sound Level Meter and Octave Band Analyzer

The HRT maintains multipurpose Type I sound level meters and octave band analyzers, which can also be operated as sound intensity analyzers for identifying noise sources and determining engineering controls. In addition, this equipment includes a building acoustics system for measuring noise decay and determining the reverberation characteristics for a given room. Based on the noise decay data, calculations can be performed to estimate potential noise reduction if absorptive materials are applied to room surfaces, such as the walls and ceiling.

Specialized Noise Dosimeters

The HRT maintains super-duty noise dosimeters that are contained in a sealed, waterproof, intrinsically safe metal housing. The dosimeters have no controls or displays, which eliminates the possibility of tampering or damage by the individual wearing the monitor. The dosimeters are programmed and controlled using a remote control unit or personal computer. The remote control unit can also be operated as an additional dosimeter. Data is transferred from the dosimeter via an infrared port on the dosimeter housing.